FIITJEE Students From Classroom / Integrated School Programs have secured to 2 Zonal, 6 State & 18 City Topper Ranks. 33 in Top 100, 78 in Top 200 and 205 in Top 500 All India Ranks bagged by FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced)-2016

FULL TEST – II

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 243**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 06)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (07 – 09) contains 3 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

   Section-A (10 – 14) contains 2 paragraphs. Based upon paragraph, 2 and 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.
2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct matching. No marks will be given for any wrong matching in any question. There is no negative marking.
3. **Section-C (01 – 05)** contains 5 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

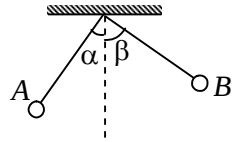
Straight Objective Type

This section contains 6 multiple choice questions numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. A particle is kept at rest at the top of a sphere of diameter 42 m. When disturbed slightly, it slides down. At what height *h* from the bottom, the particle will leave the sphere
(A) 14 m (B) 16 m
(C) 35 m (D) 70 m

2. A spherical object of mass 1 kg and radius 1 m is falling vertically downward inside a viscous liquid in a gravity free space. At a certain instant the velocity of the sphere is 2 m/s. If the coefficient of viscosity of the liquid is $\frac{1}{18\pi}$ N-s/m^2, then velocity of ball will become 0.5 m/s after a time
(A) ln 4 s (B) 2 ln 4 s
(C) 3 ln 4 s (D) 2 ln 2 s

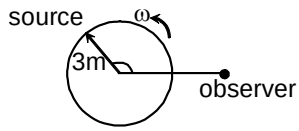
3. Two identical simple pendulums *A* and *B* are fixed at same point. They are displaced by an angle α and β (α and β are very small and $\beta > \alpha$) and released from rest. Find the time after which *B* reaches its initial position for the first time. Collisions are elastic and length of the strings is *l*.



(A) $\pi\sqrt{\frac{l}{g}}$ (B) $2\pi\sqrt{\frac{l}{g}}$
(C) $\frac{\pi\beta}{\alpha}\sqrt{\frac{l}{g}}$ (D) $\frac{2\pi\beta}{\alpha}\sqrt{\frac{l}{g}}$

Space for Rough work

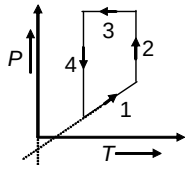
4. A source is moving on a circle of radius 3 m with constant angular velocity ω = 5 rad/s. If the observer is at a distance 5 m from the centre of circle, the time interval between maximum and minimum frequency received by the observer is



(A) $\frac{\pi}{5}$ (B) $\frac{2}{5}\cos^{-1}\left(\frac{3}{4}\right)$

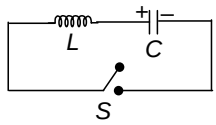
(C) $\frac{2}{5}\cos^{-1}\left(\frac{3}{5}\right)$ (D) $\frac{2}{5}\sin^{-1}\left(\frac{3}{5}\right)$

5. In the cyclic process 1-2-3-4 shown in figure for a monoatomic gas, the volume of the gas in process 1.



(A) increases
(B) decreases
(C) constant
(D) first increases then decreases

6. A capacitor is charged to a potential of V_0. It is connected with an inductor through a switch S. The switch is closed at time t = 0. Which of the following statement is correct.



(A) maximum current in the circuit is $V_0\sqrt{LC}$

(B) potential across capacitor becomes zero for the first time at $t = \pi\sqrt{LC}$

(C) energy stored in the inductor at time $\frac{\pi}{2}\sqrt{LC}$ is $\frac{1}{4}CV_0^2$

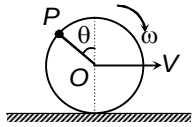
(D) maximum energy stored in the inductor is $\frac{1}{2}CV_0^2$.

Space for Rough work

Multiple Correct Choice Type

This section contains 3 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

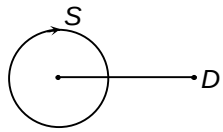
7. A disc of radius *R* rolls on a horizontal surface with linear velocity *V* and angular velocity ω. There is a point *P* on circumference of disc at angle θ with upward vertical diameter measured anticlockwise (see figure), which has a vertical velocity. Here θ is equal to



(A) $\pi + \sin^{-1}\frac{V}{R\omega}$ (B) $\frac{\pi}{2} - \sin^{-1}\frac{V}{R\omega}$

(C) $\pi - \cos^{-1}\frac{V}{R\omega}$ (D) $\pi + \cos^{-1}\frac{V}{R\omega}$

8. A source of sound moves along a circle of radius 2 m with constant angular velocity 40 rad/s. Frequency of the source is 300 Hz. A detector is kept at some distance from the circle in the same plane of the circle (as shown in figure). Which of the following is not the possible value of frequency registered by the detector? (Speed of sound = 320 m/s)



(A) 250 Hz (B) 360 Hz

(C) 410 Hz (D) 220 Hz

9. An electron in H-atom jumps from second excited state to first excited state and then from first excited to ground state. Let the ratio of wavelength, momentum and energy of photons emitted in these two cases be *a, b* and *c* respectively. Then

(A) $c = \frac{1}{a}$ (B) $a = \frac{9}{4}$

(C) $b = \frac{5}{27}$ (D) $c = \frac{5}{27}$

Space for Rough work

Comprehension Type

This section contains 2 paragraphs. Based upon one of paragraph 2 multiple choice questions and based on the other paragraphs 3 multiple choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 10 and 11

Four satellites *A*, *B*, *C* and *D* are orbiting around a spherical planet of uniform density. The orbits of all the satellites are coplanar. Orbits of satellites *A* and *B* are circular where radius of orbit *A* is greater than diameter of orbit *B*. Orbit *C* is touching orbits *A* and *B*. While orbit *D* is touching orbit *A* but its nearest distance from the planet is least among all four orbits. All the satellites have same mass and these satellite can be considered as particles. There is no gravitational force between the satellite and also there is no chance of collision between the satellites. Consider gravitational potential energy to be zero under infinite separation.

10. In which orbit can the gravitational potential energy have least possible value?
    (A) A (B) B
    (C) C (D) D

11. In which orbit does the angular momentum have largest magnitude?
    (A) A (B) B
    (C) C (D) D

Space for Rough work

Paragraph for Question Nos. 12 and 14

A thin non-conducting ring of mass *m*, radius *a* carrying a charge *q* can rotate freely about its own axis which is vertical. At the initial moment the ring was at rest in horizontal position and no magnetic field was present. At instant $t = 0$ A uniform magnetic field is switched on which is vertically downward and increases with time according to the law $B = B_0 t$. Neglecting magnetism induced due to rotational motion of ring.

12. The magnitude of induced emf will be

(A) $\pi a^2 B_0$ (B) $2a^2 B_0$

(C) zero (D) $\frac{1}{2}\pi a^2 B_0$

13. The magnitude of an electric field on the surface of the ring is

(A) aB_0 (B) $2aB_0$

(C) $\frac{1}{2}aB_0$ (D) zero

14. Find instantaneous power developed by electric force acting on the ring at $t = 1$ sec.

(A) $\frac{2q^2 B_0^2 a^2}{14m}$ (B) $\frac{q^2 B_0^2 a^2}{8m}$

(C) $\frac{3q^2 B_0^{\,2} a^2}{m}$ (D) $\frac{q^2 B_0^{\,2} a^2}{4m}$

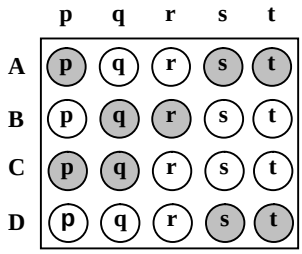
Space for Rough work

SECTION – B

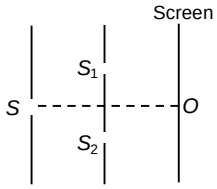
(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



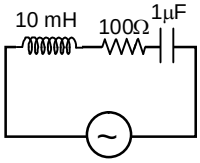
1. Figure shows a set-up to perform Young's double slit experiment. A monochromatic source of light is placed at S. S_1 and S_2 act as coherent sources and interference pattern is obtained on the screen. Match Column-I with Column-II in regard to interference in Young's double slit experiment:



	Column I		Column II
(A)	A thin transparent plate is placed in front of S_1. Assuming negligible absorption by the plate	(p)	Interference fringes disappear
(B)	S_1 is closed	(q)	There is uniform illumination on a large part of the screen
(C)	A thin transparent plate is placed in front of S_2.	(r)	The zero order fringe will not form at O
(D)	S is removed and two real sources emitting light of same wavelength are placed at S_1 and S_2.	(s)	Central maxima is formed below O
		(t)	

Space for Rough work

2. Referring to the given circuit, match Column-I with Column-II.



V = 10 sin ωt (Volt)

	Column I		Column II
(A)	For ω = 8000 rad/s	(p)	Peak current in the circuit is less than 0.1 A
(B)	For ω = 10000 rad/s	(q)	Voltage across the combination and the current are in same phase
(C)	For ω = 10500 rad/s	(r)	Voltage across the combination leads the current
(D)	For ω = 10000 rad/s, if R = 50 Ω instead of 100 Ω	(s)	Current through the circuit leads the voltage across it.
		(t)	

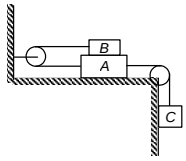
SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. The lengths of sides of a cuboid are a, $2a$ and $3a$. If the relative percentage error in the measurement of a is 1%, then find the relative percentage error in the measurement of volume of cube (in percentage)….

2. For the arrangement shown in figure, the masses of the blocks are m_A = 2 kg, m_B = 1 kg, m_C = 1 kg. Find the frictional force between the blocks A and B (in Newton) (coefficient of friction between any two surfaces is 0.5)



3. Suppose potential energy between particle A and B at separation r is given by $U = k \ln r$, where k is constant. For such hypothetical system, find the ratio of energy difference between energy levels (n = 1 and n = 2) and (n = 2 and n = 4) (The energy of a particle in n^{th} level is $-13.6/n^2$)

4. A uniform disc of radius R having charge Q distributed uniformly all over its surface is placed on a smooth horizontal surface. A magnetic field, $B = kxt^2$, where k is a constant, x is the distance (in metre) from the centre of the disc and t is the time (in second), is switched on perpendicular to the plane of the disc. Find the torque (in N-m) acting on the disc after 15 sec. (Take $4kQ$ = 1 S.I. unit and R = 1m) is

5. The hydrogen atom in its ground state are excited by means of monochromatic radiation of wavelength 1023 Å. Find the number of lines which can be seen in the resulting spectrum…

Space for Rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains 6 multiple choice questions numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. Which of these reaction is correct:
   (A) $Br^- + Cl_2 \longrightarrow Cl^- + Br_2$
   (B) $Mohr's\ salt \xrightarrow{Zn/NaOH} N_2\,(g)$
   (C) $K_2Cr_2O_7 solution \xrightarrow{SO_3}$ Green colour
   (D) $FeCl_2 \xrightarrow{NaOH} (ppt.) \xrightarrow{Excess\ NaOH}$ Soluble complex

2. Potash alum contains the metals like potassium and aluminium. What are the metal ions present in chrome alum?
   (A) Chromium and aluminium only
   (B) Iron and aluminium
   (C) Potassium and chromium only
   (D) Potassium, chromium and aluminium

3. For a gaseous reaction, the rate of reaction is expressed in terms of $\frac{dP}{dt}$ instead of $\frac{dC}{dt}$ or $\frac{dn}{dt}$, where C is concentration and n is the number of moles. Hence, the relation between expression is:
   (A) $\frac{dC}{dt} = \frac{dn}{dt} = \frac{V}{RT}\frac{dP}{dt}$
   (B) $\frac{dC}{dt} = \frac{1}{V}\frac{dn}{dt} = \frac{1}{RT}\frac{dP}{dt}$
   (C) $\frac{dP}{dt} = \frac{dn}{dt} = \frac{dC}{dt}$
   (D) None of these

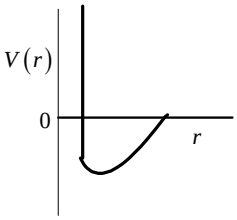
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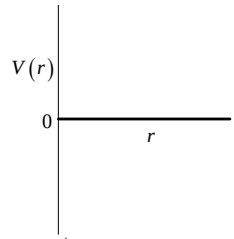
4. An excited electron of H-atoms emits of photon of wavelength λ and returns in the ground state, the principal quantum number of excited state is given by:

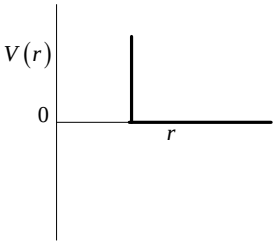
(A) $\sqrt{\lambda R(\lambda R-1)}$ (B) $\sqrt{\dfrac{\lambda R}{(\lambda R-1)}}$

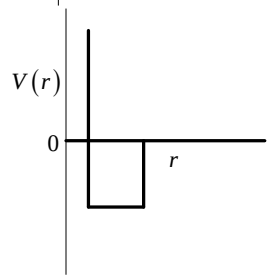
(C) $\sqrt{\dfrac{1}{\lambda R(\lambda R-1)}}$ (D) $\sqrt{\dfrac{(\lambda R-1)}{\lambda R}}$

5. One mole of a monatomic real gas satisfies the equation $p(V-b)=RT$ where b is a constant. The relationship of interatomic potential $V(r)$ and interatomic distance r for the gas is given by

(A) 

(B) 

(C) 

(D) 

6. Which of the following will not undergo aldol condensation?

(A) trimethyl acetaldehyde (B) formaldehyde

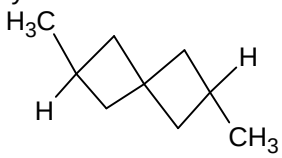
(C) benzaldehyde (D) benzophenone

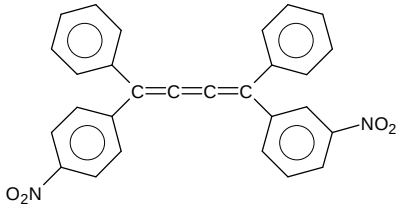
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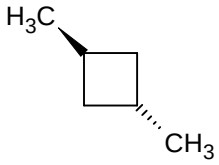
Multiple Correct Choice Type

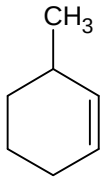
This section contains 3 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

7. Identify the correct relationship:
   (A) $\left[Cu\left(NH_3\right)_4\right]^{2+}$ with Hybridization of $Cu = dsp^2$
   (B) $Fe\left(CO\right)_x$ with $x = 5$
   (C) Ni^{2+} Complexes with dsp^2 hybridization and dimagnetic nature
   (D) Cr^{3+} Complexes with Paramagnetic nature

8. Identify the chiral molecule/molecules:
   (A) 

   (B) 

   (C) 

   (D) 

9. The correct statement/statements is/are:
   (A) the distance between two octahedral voids of a cubic close packing of atoms with radius r is 2r
   (B) the distance between two tetrahedral voids of a cubic close packing of atoms with radius r is $\sqrt{6} \times r$ at the body diagonal axes
   (C) the distance between tetrahedral and octahedral void is $\sqrt{\frac{3}{2}} \times r$, where is the radius of atom forming the close packing
   (D) the distance between the surface of atoms fcc arrangement along an edge is $2\left(\sqrt{2}-1\right)r$

Space for Rough work

Comprehension Type

This section contains 2 paragraphs. Based upon one of paragraph 2 multiple choice questions and based on the other paragraphs 3 multiple choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 10 and 11

The cloud consist of charged particles of water dispersed in air. Some of them are +vely charged and other are -vely charged. When +vely charged clouds come closer they have lightening and thundering whereas - ve and +ve charged colloids come closer they cause heavy rain due to the aggregation of minute particles. It is possible to cause artificial rain by throwing electrified sand or AgI from aeroplane and thus coagulation the mist hanging in air.

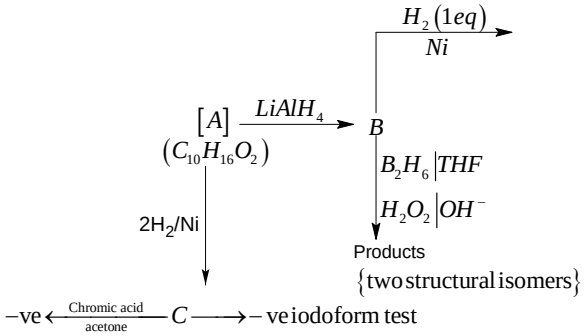
10. When excess of $AgNO_3$ is treated with KI solution, AgI forms.
    (A) + vely charged solution
    (B) - vely charged solution
    (C) Neutral solution
    (D) True solution

11. Electrical chimney are made on the principal of
    (A) Electro-osmosis
    (B) Electrophoresis
    (C) Coagulation
    (D) All of these

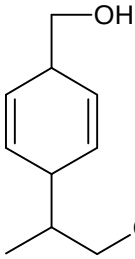
Space for Rough work

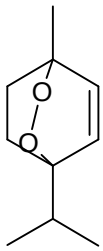
Paragraph for Question Nos. 12 and 14

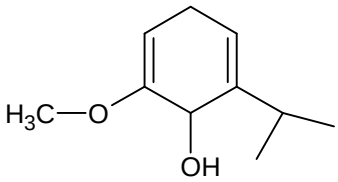
Compound (A), which can be distilled in high vacuum, gives positive test with Br_2 / H_2O, but does not react with Na metal in ethereal solution, but compound (B) *and* (C) gives $H_2\,(g)$ with sodium metal. Compound (A) undergoes following sequence of reactions-

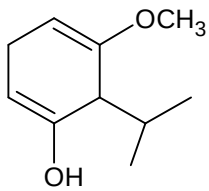


12. Compound 'A' should be:

(A)


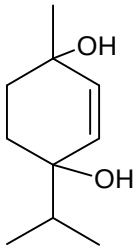
(B)


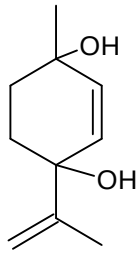
(C)


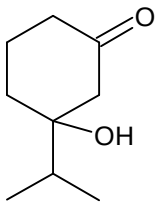
(D)


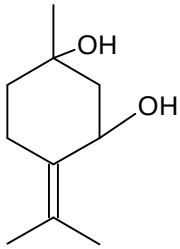
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13. Compound 'B'

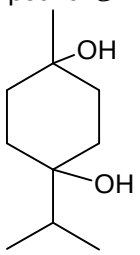
(A) 

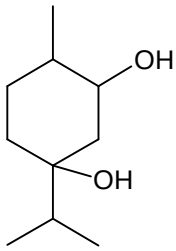
(B) 

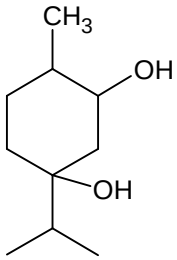
(C) 

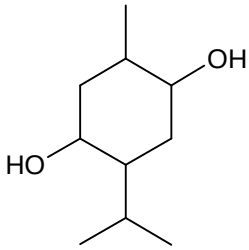
(D) 

14. Compound 'C'

(A) 

(B) 

(C) 

(D) 

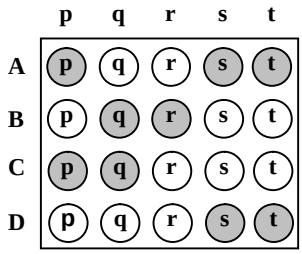
Space for Rough work

SECTION – B

(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Match the following metal given in column-I with the appropriate metal exaction process listed with column- II

Column I		Column II	
(A)	*Fe*	(p)	Cyanide Process
(B)	*Ag*	(q)	Leaching
(C)	*Pb*	(r)	Carbon reduction
(D)	*Cu*	(s)	Self reduction
		(t)	Electrolysis

2.

Column I		Column II	
(A)	$[Fe(NH_3)_6]^{3+}$ $[Fe(H_2O)_6]^{3+}$ $[Fe(H_2O)_6]^{2+}$ (I) (II) (III)	(p)	$I > II > III$ (decreasing order of stability for complexes
(B)	$[NiCl_4]^{2-}$ $[PdCl_4]^{2-}$ $[PtCl_4]^{2-}$ (I) (II) (III)	(q)	At least one complex shows stereoisomerism
(C)	$[V(CO)_6]^{-}$, $[V(NH_3)_6]^{3+}$, $[V(NH_3)_4Cl_2]^{+}$ (I) (II) (III)	(r)	All complexes has $EAN \geq 36$
(D)	$[Ni(EDTA)]^{2-}$, $[Ni(dien)_2]^{2+}$, $[Ni(en)_3]^{2+}$ (I) (II) (III)	(s)	At least two complexes are d^2sp^3 *or* dsp^2 hybridised
		(t)	At least one complex is paramagnetic in nature

Space for Rough work

SECTION – C

(One Integer Value Correct Type)

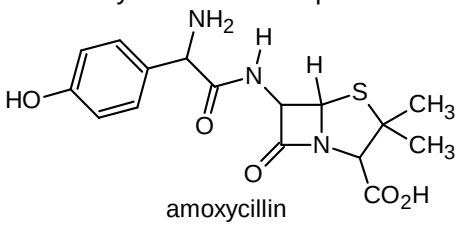
This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. A saturated solution of spraingly soluble salt MCl_2 has a vapour pressure of 31.78 mm of Hg at $30°C$ while pure water exerts a pressure of 31.82 mm of Hg at the same temperature. The solubility product is $x \times 10^{-x}$. The value of x is

2. The number of unpaired e^- in first excited state of O_2 is

3. The number of geometrical isomers of the octahedral complex of the type [Mabcdef] is $n \times 3.$. The value of n is

4. How many double bond equivalents does amoxicillin (shown below) possesses?



amoxycillin

5. $(i)\ FeSO_4 + KCN\,(excess) \longrightarrow (X)$

$(X) \xrightarrow{conc.\,H_2SO_4} (A)\,gas$

$(X) \xrightarrow{dil\,H_2SO_4} (B)\,gas$

$(ii)\ CaC_2 + N_2 \xrightarrow{1100°C} (Y) + carbon$

Find the sum of the number of π bonds present in (A), (B) and (Y):

Space for Rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains 6 multiple choice questions numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. $A(-2,0), B(0,4)$. The point M on line y = x for which perimeter of ΔAMB is least is
   (A) (0, 0) (B) $(-1,-1)$
   (C) $\left(\frac{1}{2},\frac{1}{2}\right)$ (D) none of these

2. Equation of common tangent to $y^2=8x$ and $xy=1$ may be
   (A) $3y=9x+2$ (B) $y=2x+1$
   (C) $2y=x+8$ (D) none of these

3. Let $S=\sum_{r=1}^{117}\frac{1}{2[\sqrt{r}]+1}$ where [.] denotes the greatest integer function. The value of S is
   (A) $\frac{69}{7}$ (B) $\frac{206}{21}$
   (C) $\frac{76}{7}$ (D) $\frac{227}{21}$

4. Minimum value of function $f(x)=x^3(x^3+1)(x^3+2)(x^3+3): x\in R$, is
   (A)–2 (B) –1
   (C) 1 (D) none

Space for Rough work

5. The hands of a clock has lengths 3 & 4 units. The distance between their tips when the distance is increasing most rapidly is

(A) 2 (B) $\sqrt{7}$

(C) $\sqrt{3}$ (D) 5

6. $\overline{OA} = 2i + j + 2k$ is rotated about O (origin) by 90° such that it crosses the positive y axis during the rotation. If A goes to B then $\overline{OB}$ =

(A) $\frac{1}{\sqrt{2}} i - 2\sqrt{2} j + \frac{1}{\sqrt{2}} k$ (B) $-\frac{1}{\sqrt{2}} i + 2\sqrt{2} j - \frac{1}{\sqrt{2}} k$

(C) $2\sqrt{2} i - \frac{1}{\sqrt{2}} j - \frac{1}{\sqrt{2}} k$ (D) $-2\sqrt{2} i + \frac{1}{\sqrt{2}} j + \frac{1}{\sqrt{2}} k$

Multiple Correct Choice Type

This section contains 3 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

7. In ΔABC, if $A = 45°, B = 75°$ then

(A) $a + \sqrt{2} c = 2b$ (B) $\sqrt{3} a = \sqrt{2} c$

(C) $a = \left(\sqrt{3} - 1\right) b$ (D) $3a = 5b$

8. Equation of plane through $A(1, 0, 0), B(0, 1, 0)$ and making an angle $\frac{\pi}{4}$ with the plane $x + y = 3$ can be

(A) $x - y + 2z = 1$ (B) $x + y + \sqrt{2} z = 1$

(C) $x + y - \sqrt{2} z = 1$ (D) $x - y - 2z = 1$

Space for Rough work

9. If the area bounded by the curve $y = x - x^2$ and the line y = mx equals $\frac{9}{2}$. Then the value of q m is

(A) -4 (B) -2

(C) 2 (D) 4

Comprehension Type

This section contains 2 paragraphs. Based upon one of paragraph 2 multiple choice questions and based on the other paragraphs 3 multiple choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 10 and 11

A box has 10 coins. Five have heads on both sides. Three have tails on both sides; two are fair. A coin is chosen at random and tossed.

10. Probability that head appears is

(A) $\frac{4}{5}$ (B) $\frac{2}{5}$

(C) $\frac{3}{5}$ (D) $\frac{1}{5}$

11. Probability that the coin is fair given that head has appeared is

(A) $\frac{5}{6}$ (B) $\frac{1}{6}$

(C) $\frac{3}{8}$ (D) $\frac{5}{8}$

Space for Rough work

Paragraph for Question Nos. 12 and 14

ABC is a triangle. Complex numbers z_1, z_2, z_3 are represented by A, B, C respectively. Which are in anticlock wise order. Let $|z_1| = |z_2| = |z_3|$ and arg $\frac{z_3 - z_1}{z_2 - z_1} = \frac{\pi}{6}$

12. $|z_2 - z_3| =$

(A) $\frac{1}{2}|z_1 - z_3| \cos ec\left(\frac{1}{2}\arg\frac{z_1 - z_2}{z_3 - z_2}\right)$ (B) $\frac{1}{2}|z_1 - z_3| \text{cosec}\left(\arg\frac{z_1 - z_2}{z_3 - z_2}\right)$

(C) $\frac{1}{2}|z_1 - z_3| \sin\left(\frac{1}{2}\arg\frac{z_1 - z_2}{z_3 - z_2}\right)$ (D) $\frac{1}{2}|z_1 - z_3| \sin\left(\arg\frac{z_1 - z_2}{z_3 - z_2}\right)$

13. Orthocentre H of ΔABC denotes the complex no.

(A) $i(z_1 + z_2 + z_3)$ (B) $\frac{z_1 + iz_2 + z_3}{2}$

(C) $z_1 + z_2 + z_3$ (D) $\frac{iz_1 + z_2 + z_3}{3}$

14. If $T(z)$ is any point such that HBTC is a parallelogram then $|z - z_1| =$

(A) $2|z_1 - z_3|$ (B) $2|z_2 - z_3|$

(C) $2|z_1 - z_2|$ (D) none

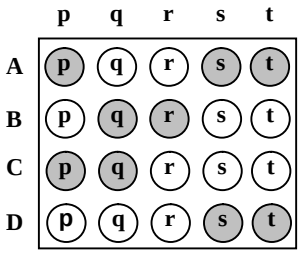
Space for Rough work

SECTION – B

(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1.

	Column I		Column II
(A)	The coordinates of the point on the parabola $y = x^2 + 7x + 2$, which is nearest to the straight line y = 3x – 3 are	(p)	(2,1)
(B)	y = x + 2 is a tangent to the parabola $y^2 = 8x$. The point on this line, the other tangent from which is perpendicular to this tangent is	(q)	(–2, 0)
(C)	The point on the ellipse $x^2 + 2y^2 = 6$ whose distance from the line x + y = 7 is least is	(r)	$\left(2, \frac{2}{\sqrt{3}}\right)$
(D)	The foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ are S and S'. P is a point on the ellipse whose eccentric angle is $\pi/3$. The incentre of the triangle SPS' is	(s)	(–2,–8)
		(t)	(2, 2)

Space for Rough work

2.

	Column I		Column II
(A)	If sin $\theta = 3\sin(\theta + 2\alpha)$, then the value of $\tan(\theta + \alpha) + 2\tan\alpha$ is	(p)	0
(B)	If p sin θ + q cos θ = a and p cos θ − q sin θ = b then $\frac{p+a}{q+b} + \frac{q-b}{p-a} + 1$ is equal to	(q)	1
(C)	The value of the expression $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{10\pi}{7} - \sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}$ is	(r)	sec θ
(D)	If $\sec\theta + \tan\theta = 1$, then one root of the equation $(a-2b+c)x^2 + (b-2c+a)x + (c-2a+b) = 0$ is	(s)	$-\frac{1}{4}$
		(t)	-1/2

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. In equilateral triangle ABC of side p, a square is inscribed in its incircle. If area of the square is $\frac{p^2}{k}$ then value of *k* is

2. *f* is a function such that $f(x) + 2f(1-x) = x^2 + 1$. Value of $f(3)$ is

3. Degree of differential equation of family of ellipses of same size having minor axis along a given line is

4. In ΔABC, $\tan A : \tan B : \tan C \equiv 1 : 2 : 3$. If $\sin A : \sin B : \sin C \equiv \sqrt{5} : 2\sqrt{2} : \lambda$, then $\lambda =$

5. Coefficient of x^6 in the expansion of $(1 + x + x^2 + x^3)^{-3}$ is

Space for Rough work